

2015 年真题参考答案

一、选择题

(1) D. (2) B. (3) A. (4) C. (5) D. (6) B. (7) D. (8) A.

二、填空题

(9) 48. (10) $n(n-1)(\ln 2)^{n-2}$. (11) 2. (12) $2e^x + e^{-2x}$. (13) $-\frac{1}{3}dx - \frac{2}{3}dy$. (14) 21.

三、解答题

(15) $a = -1$, $b = -\frac{1}{2}$, $k = -\frac{1}{3}$.

(16) $\frac{8}{\pi}$.

(17) 极小值 $f(0, -1) = -1$.

(18) $\frac{\pi}{4} - \frac{2}{5}$.

(19) $f(x)$ 在 $(-\infty, +\infty)$ 上共有两个零点.

(20) 30 min.

(21) 证明略.

(22) (I) $a = 0$.

$$(II) \mathbf{X} = \begin{pmatrix} 3 & 1 & -2 \\ 1 & 1 & -1 \\ 2 & 1 & -1 \end{pmatrix}.$$

(23) (I) $a = 4$, $b = 5$.

$$(II) \mathbf{P} = \begin{pmatrix} 1 & 2 & 3 \\ 1 & 1 & 0 \\ -1 & 0 & -1 \end{pmatrix}, \mathbf{P}^{-1}\mathbf{A}\mathbf{P} = \begin{pmatrix} 5 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$